Sitong Wang

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Research Interests

Human-AI Interaction, Multimodal Generative AI, Agentic AI

Education

2020-09/2026 **Ph.D. in Computer Science**, Columbia University
Advised by: Lydia B. Chilton

M.S. in Computer Science, Columbia University
Advised by: Lydia B. Chilton; *Track:* Machine Learning

2015-2020 **B.S. in Electrical Engineering**, University of Cincinnati
B.E. in Electrical Engineering and Automation, Chongqing University

Awards and Honors

Aug 2023 **Code Quality Jam Best Robustness Award**
Adobe Research

Sep 2022 **Greenwoods Fellowship Receipent**
Columbia University

Jun 2021 **Google CS Research Mentorship Program Class of 2021**
Google Research

Nov 2017&2018 **National Scholarship Recipient**
Chinese Ministry of Education

Publications

CONFERENCE AND JOURNAL PAPERS

2026 [C.13] **Rewriting Video: Text-Driven Reauthoring of Video Footage.**
Sitong Wang, Anh Truong, Lydia B. Chilton, Dingzeyu Li.
In ACM Conference on Intelligent User Interfaces (IUI'26).

2025 [C.12] **Copying style, Extracting value: Illustrators' Perception of AI Style Transfer and its Impact on Creative Labor.**
Julien Porquet, Sitong Wang, Lydia B. Chilton.
In ACM Conference on Human Factors in Computing Systems (CHI'25).

2025 [C.11] **DynEx: Dynamic Code Synthesis with Structured Design Exploration for Accelerated Exploratory Programming.**
Jenny Ma, Karthik Sreedhar, Vivian Liu, Pedro A. Perez, Sitong Wang, Riya Sahni, Lydia B. Chilton.
In ACM Conference on Human Factors in Computing Systems (CHI'25).
Best Paper Honorable Mention Award (Top 5%)

2024 [C.10] **MoodSmith: Enabling Mood-Consistent Multimedia for AI-Generated Advocacy Campaigns.**
Samia Menon, Sitong Wang, Lydia B. Chilton.
In International Conference on Computational Creativity (ICCC'24).

2024 [C.9] **PodReels: Human-AI Co-Creation of Video Podcast Teasers.**
Sitong Wang, Zheng Ning, Anh Truong, Mira Dontcheva, Dingzeyu Li, Lydia B. Chilton.
In ACM Designing Interactive Systems Conference (DIS'24).

2024 [C.8] **ReelFramer: Human-AI Co-Creation for News-to-Video Translation.**
Sitong Wang, Samia Menon, Tao Long, Keren Henderson, Dingzeyu Li, Kevin Crowston, Mark Hansen, Jeffrey V. Nickerson, Lydia B. Chilton.
In ACM Conference on Human Factors in Computing Systems (CHI'24).

2024 [C.7] **A Design Space for Intelligent and Interactive Writing Assistants.**
Mina Lee, Katy Ilonka Gero, John Joon Young Chung, [et al., including Sitong Wang].
In ACM Conference on Human Factors in Computing Systems (CHI'24).

2023 [C.6] **Tweetorial Hooks: Generative AI Tools to Motivate Science on Social Media.**
Tao Long, Dorothy Zhang, Grace Li, Batool Taraif, Samia Menon, Kynnedy Simone Smith, Sitong Wang, Katy Ilonka Gero, Lydia B. Chilton.
In International Conference on Computational Creativity (ICCC'23).

2023 [C.5] **PopBlends: Strategies for Conceptual Blending with Large Language Models.**
Sitong Wang, Savvas Petridis, Taeahn Kwon, Xiaojuan Ma, Lydia B. Chilton.
In ACM Conference on Human Factors in Computing Systems (CHI'23).

2022 [C.4] **Metaphoraction: Support Gesture-based Interaction Design with Metaphorical Meanings.**
Zhida Sun, Sitong Wang, Chengzhong Liu, Xiaojuan Ma.
In ACM Transactions on Computer-Human Interaction (TOCHI'22).

2022 [C.3] **Know it to Defeat it: Exploring Health Rumor Characteristics and Debunking Efforts on Chinese Social Media during COVID-19.**
Wenjie Yang, Sitong Wang*, Zhenhui Peng*, Chuhan Shi, Diyi Yang, Xiaojuan Ma.
In AAAI International Conference on Web and Social Media (ICWSM'22).

2021 [C.2] **MetaMap: Supporting Visual Metaphor Ideation through Multi-dimensional Example-based Exploration.**
Youwen Kang*, Zhida Sun*, Sitong Wang, Zeyu Huang, Ziming Wu, Xiaojuan Ma.
In ACM Conference on Human Factors in Computing Systems (CHI'21).

2020 [C.1] **"A Postcard from Your Food Journey in the Past": Promoting Self-Reflection on Social Food Posting.**
Zhida Sun, Sitong Wang, Wenjie Yang, Onur Yürüten, Chuhan Shi, Xiaojuan Ma.
In ACM Designing Interactive Systems Conference (DIS'20).

UNDER SUBMISSION

2025 [M.5] **Schemex: Interactively Discovering Structural Abstractions from Examples.**
Sitong Wang, Samia Menon, Dingzeyu Li, Xiaojuan Ma, Richard Zemel, Lydia B. Chilton.
Under submission.

2025 [M.4] **DoubleAgents: Interactive Simulations for Alignment in Agentic AI.**
Tao Long*, Xuanming Zhang*, Sitong Wang, Zhou Yu, Lydia B. Chilton.
Under submission.

2025 [M.3] **JumpStarter: Human-AI Planning with Task-Structured Context Curation.**
Xuanming Zhang*, Sitong Wang*, Jenny Ma, Alyssa Hwang, Andrew Head, Zhou Yu, Lydia B. Chilton.
Under submission.

2025 [M.2] **Self-Rewarding Reinforcement Learning for Machine Translation.**
Wenjie Yang, Mao Zheng, Mingyang Song, Zheng Li, Sitong Wang.
Under submission.

2025 [M.1] **Fostering Engagement in Creative Thinking.** *Chapter in* **Engagement in Human-Centered, Intelligent Computing**, Xiaojuan Ma (Ed.).
Zhida Sun, Sitong Wang.
Book under submission to CRC Press.

WORKSHOP PAPERS

2025 [W.2] **Schemex: Discovering Design Patterns from Examples with Iterative Abstraction and Refinement.**
Sitong Wang, Lydia B. Chilton.
In Tools for Thought: Research and Design for Understanding, Protecting, and Augmenting Human Cognition with Generative AI (CHI'25 workshop).

2023 [W.1] **Writing with Generative AI: Multi-modal and Multi-dimensional Tools for Journalists.**
Sitong Wang, Lydia B. Chilton, Jeffery V. Nickerson.
In the Second Workshop on Intelligent and Interactive Writing Assistants (CHI'23 workshop).

Research Experience

2020-NOW **Computational Design Lab**, Columbia University
Advised by: Lydia B. Chilton
Developed an AI-driven interactive system to help journalists transform print articles into social media reels [C.8]; Designed an automated pipeline that combines knowledge extraction with large language models to blend a given topic with pop culture domains [C.5].

Summer 2025 **STORIE Lab**, Adobe Research
Advised by: Dingzeyu Li and Anh Truong
Researched generative video reconstruction algorithms and agentic workflows to develop adaptive, text-driven AI systems that support new forms of creative video production [C.13].

Summer 2023 **Interactive Creativity Lab**, Adobe Research
Advised by: Dingzeyu Li, Anh Truong, Mira Dontcheva
Derived design principles for video podcast teasers through interviews, designed a human-AI co-creative workflow, and implemented a Premiere plugin that integrates with creators' existing workflows, significantly easing their mental burden and enhancing efficiency [C.9].

Summer 2020 **Human-Computer Interaction Lab**, Hong Kong University of Science and Technology
Advised by: Xiaojuan Ma, Diyi Yang
Conducted a literature survey on online health rumors, contributing to both quantitative and qualitative analyses [C.3]; Analyzed the visual metaphor creation process through interviews, contributed to user studies and framework design for visual metaphor ideation [C.2].

Summer 2019 **Human-Computer Interaction Lab**, Hong Kong University of Science and Technology
Advised by: Xiaojuan Ma
Helped propose an automated postcard generation system based on social media food posts to promote self-reflection on health [C.1]; Conducted a literature survey on mobile gestures and analyzed user interpretations from an MTurk study to derive design principles [C.4].

Teaching Experience

Fall 2023 **COMS 6998 Designing for Generative AI**
Teaching Assistant, Columbia University

Fall 2022 **COMS 6998 Advanced Web Design Studio**
Teaching Assistant, Columbia University

Mentoring Experience

2023-NOW Samia Menon (Columbia CS undergraduate student) EECS PhD student at the University of California, Berkeley

2024-2025 Jenny Ma (Columbia CS masters student) CS PhD student at Columbia University

2024-2025 Karthik Sreedhar (Columbia CS masters student) CS PhD student at Columbia University

2023-2024 Grace Li (Barnard CS undergraduate student) CS PhD student at the University of Chicago

Community Service

2025 **Workshop Organizer.** Facilitating Longitudinal Interaction Studies of AI Systems. The ACM Symposium on User Interface Software and Technology (UIST)

2025-2026 **Program Committee Member.** The ACM conference on Creativity & Cognition (C&C)

2023-2026 **Conference Reviewer.** SIGGRAPH (2025-2026), ACL (2025), CHI (2023-2026), UIST (2024-2025), DIS (2024-2026), CSCW (2023-2024), ICWSM (2023-2024), C&C (2024), TEI (2024)

2023-2026 **Journal Reviewer.** IJHCI (2024, 2026), TIST (2024), TOCHI (2023), HCI Journal (2026)

2022-2024 **Volunteer.** Columbia CS PhD Pre-submission Application Review Program

Professional Experience

Sep 2022-NOW **Computational Design Lab**, Columbia University
Graduate Research Assistant

Summer 2025 **STORIE Lab**, Adobe Research
Research Scientist/Engineer Intern

Summer 2023 **Interactive Creativity Lab**, Adobe Research
Research Scientist/Engineer Intern

Spring 2022 **Alexa Mobile Application Development Group**, Amazon
Software Engineer

Summer 2021 **Alexa Mobile Application Development Group**, Amazon
Software Engineer Intern

Spring 2018 **Big Data Tool Development Group**, CISDI Engineering
User Interface Designer Intern